$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & \frac{1}{2}\left(-3\,Y_1\,k^2-3\,\overset{(2)}{\kappa}_3\right) & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & 0 & 0 & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & 0 & 0 & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -k^2\,\overset{(4)}{\kappa}_2-\overset{(2)}{\kappa}_3 & \frac{k^2\,\overset{(4)}{\kappa}_2+\overset{(2)}{\kappa}_3}{\sqrt{2}} \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{k^2\,\overset{(4)}{\kappa}_2+\overset{(2)}{\kappa}_3}{\sqrt{2}} & \frac{1}{2}\left(-k^2\,\overset{(4)}{\kappa}_2-\overset{(2)}{\kappa}_3\right)
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & 0 & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & \frac{1}{\mathrm{Det}(0^+)} & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & 0 & 0 & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & 0 & 0 & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{2}{3\,\mathrm{Det}(1^-)} & -\frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -\frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} & \frac{1}{3\,\mathrm{Det}(1^-)}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & 0 & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

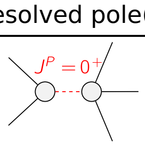
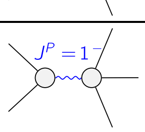
Abbreviations used in matrices

$$Y_1 == \overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2 \;\&\&\; \mathrm{Det}(0^+) == -\frac{3}{2}\left(k^2\left(\overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2\right)+\overset{(2)}{\kappa}_3\right) \;\&\&\; \mathrm{Det}(1^-) == -\frac{3}{2}\left(k^2\,\overset{(4)}{\kappa}_2+\overset{(2)}{\kappa}_3\right)$$

Lagrangian

$$\overset{(2)}{\kappa}_3\,\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\mathcal{K}^{\chi}{}_{\beta\chi}+\overset{(4)}{\kappa}_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}+\overset{(4)}{\kappa}_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{0^-}==0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta}+\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi}+\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta}+\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta}+\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta}+\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi}==$ $\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta}+\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta}+\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi}+\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta}+\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi}+\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}^{\#1\,\alpha}_{1^-}+2\,\mathcal{J}^{\#2\,\alpha}_{1^-}==0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi}+\partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta}==3\,\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}^{\#2\,\alpha\beta}_{1^+}==0$	3	$\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta}==\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}$
$\mathcal{J}^{\#1\,\alpha\beta}_{1^+}==0$	3	$\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi}==\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{2^-}==0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon}+3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta}+$ $4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta}==$ $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon}+3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta}+$ $2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
$\mathcal{J}^{\#1\,\alpha\beta}_{2^+}==0$	5	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta}+3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}+2\,\eta^{\alpha\beta}\,\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi}{}_{\chi}{}^{\delta}==2\,\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi}{}_{\chi}{}^{\delta}+3\left(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi}\right)$
Total # constraint(s):	20	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$-\frac{\overset{(2)}{\kappa}_3}{\overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2}$	$-\frac{2}{3\left(\overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2\right)}$
	3	$-\frac{\overset{(2)}{\kappa}_3}{\overset{(4)}{\kappa}_2}$	$\frac{2}{3\,\overset{(4)}{\kappa}_2}$

Resolved unitarity condition(s):	(Demonstrably impossible)